



12576 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its new sibling

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Justin Pierel (PI)	Space Telescope Science Institute
Prof. Sherry H. Suyu (CoI) (ESA Member) (CoPI)	Technical University of Munich
Conor Larison (CoI)	Space Telescope Science Institute
Dr. Jose M. Diego (CoI) (ESA Member)	Instituto de Fisica de Cantabria
Dr. Anton M. Koekemoer (CoI)	Space Telescope Science Institute
Dr. Stefan Schuldt (CoI) (ESA Member)	FINCA, University of Turku
Dr. Andrew B. Newman (CoI)	Carnegie Institution of Washington
Prof. Masamune Oguri (CoI)	Chiba University
Prof. Adi Zitrin (CoI)	Ben-Gurion University of the Negev
Dr. Patrick S. Kamienieski (CoI) (ESA Member)	Chalmers University of Technology
Dr. Martin Millon (CoI) (ESA Member)	University of Geneva, Department of Theoretical Physics
Ana Acebron (CoI) (ESA Member)	Instituto de Fisica de Cantabria
Dr. Piero Rosati (CoI) (ESA Member)	Universita degli studi di Ferrara
Dr. Pietro Bergamini (CoI) (ESA Member)	INAF - OAS Bologna Italy
Dr. Stefan Taubenberger (CoI) (ESA Member)	Max Planck Institute for Astrophysics
Sebastian Ertl (CoI) (ESA Member)	Max Planck Institute for Astrophysics
Dr. Suhail Dhawan (CoI) (ESA Member)	University of Cambridge
Dr. Louis-Gregory Strolger (CoI)	Space Telescope Science Institute
Dr. Wenlei Chen (CoI)	Oklahoma State University Main Campus
Dr. Leonidas A Moustakas (CoI)	Jet Propulsion Laboratory
Elias Mamuzic (CoI) (ESA Member)	Max Planck Institute for Astrophysics

JWST Proposal 12576 (Created: Monday, June 1, 2026, 1:00:14PM Eastern Standard Time) - Overview

Name	Institution
Dr. Massimo Pascale (CoI)	University of California - Los Angeles
Prof. Myungkook J. Jee (CoI)	Yonsei University
Mr. Sangjun Cha (not used) (CoI)	Yonsei University
Dr. Anowar J Shajib (CoI)	University of Chicago
Dr. Raoul Canameras (CoI) (ESA Member)	CNRS, Laboratoire d'Astrophysique de Marseille
Prof. Claudio Grillo (CoI) (ESA Member)	Universita di Milano
Dr. Ismael Perez-Fournon (CoI) (ESA Member)	Instituto de Astrofisica de Canarias
Dr. Saurabh W. Jha (CoI)	Rutgers the State University of New Jersey
Prof. Xiaosheng Huang (CoI)	University of San Francisco
Giovanni Granata (CoI) (ESA Member)	Portsmouth University/ICG
Prof. Brenda Louise Frye (CoI)	University of Arizona
Aadya Agrawal (CoI)	University of Illinois at Urbana - Champaign

OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
NIRCam-ToO				
	1	E1	NIRCam Imaging	(1) SN-R4
	3	E2	NIRCam Imaging	(1) SN-R4
	4	E3	NIRCam Imaging	(1) SN-R4
	5	E4	NIRCam Imaging	(1) SN-R4
	6	E5	NIRCam Imaging	(1) SN-R4
	7	E6	NIRCam Imaging	(1) SN-R4
	8	E7	NIRCam Imaging	(1) SN-R4
	9	E8	NIRCam Imaging	(1) SN-R4
NIRSpec-ToO				
	2	E1-SPEC	NIRSpec Fixed Slit Spectroscopy	(1) SN-R4
Monitoring				
	10		NIRCam Imaging	(1) SN-R4

ABSTRACT

This program will capture the reappearance of a strongly lensed Type Ia supernova (SN Ia), SN Requiem, a decade after its initial discovery. Lensed by the galaxy cluster MACS J0138-2155, the first three images of SN Requiem at $z=1.95$ were seen in a 2016 HST visit; the fourth is predicted to

appear a decade later. In 2023, this system became even more remarkable with JWST's discovery of a second lensed SN Ia in the same host galaxy, named SN Encore. Detecting SN Requiem's fourth image enables the first joint Hubble constant (H_0) measurement from two lensed SNe in one system. Using archival JWST, HST, and MUSE data, we built seven lens mass models of the cluster, incorporating both statistical and systematic uncertainties, to predict SN Requiem's reappearance. If $H_0 = 73$ km/s/Mpc, the fourth image is expected in June-Nov 2026; if $H_0 = 67.4$ km/s/Mpc, then Apr-Sept 2027. We propose two HST cycles to monitor the cluster and catch the reappearance, triggering JWST ToO imaging and spectroscopy upon detection to measure the longest-ever lensing time delay (~ 10 years). This allows a Requiem+Encore H_0 measurement with $<3\%$ uncertainty -- the most precise from lensed SNe to date -- and directly addresses the Hubble tension between early- and late-Universe values. Our program leverages HST WFC3/IR as the most efficient tool to monitor the system and detect SN Requiem, and JWST to capture the full UV-NIR photometric and spectroscopic evolution of this rare event.

OBSERVING DESCRIPTION

This is the JWST component to an HST+JWST proposal to detect the reappearance of the strongly lensed supernova (SN) Requiem at $z=2$. This is just an example follow-up program, the actual visits, filters, and exposure times will be selected to optimize follow-up given the detection phase and date of detection. We include a fixed-slit spectrum with the first visit, and then NIRCам-only spaced by 3-7 rest-frame days to fully characterize the light curve. The first visit is marked as a disruptive ToO, justified in the proposal, followed by the cadenced observations.

In addition to the triggered observations we request a single "monitoring epoch", justified in the proposal, which fills in one observing gap in the HST monitoring campaign. That gap is between November 2026-February 2027, but since we must select dates in Cycle 4 for the APT, this throws an error. We used the 2025-2026 window to calculate the program cost, and then removed this visit and requested the time for next Cycle, to be scheduled November 2026-February 2027 during HST Cycle 34. Adding hours in future cycles causes an error because this should technically be marked as a carry-over ToO since it's two HST cycles, but we cannot select both carry-over ToO and future cycle time. We therefore have removed the carry-over ToO requirement, since this program is by definition tied to the 2-cycle HST program, but this ToO should be considered "carry-over" in the sense that it can be triggered over two HST Cycles.

Proposal 12576 - Targets - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its new si...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d) Dec: -21 55 22.41 (-21.92289d) Equinox: J2000		
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO				
Generic Targets	(2)	NIRSPec-TA	RA: 01 38 2.8540 (24.5118917d) Dec: -21 55 51.17 (-21.93088d) Equinox: J2000		
	Comments: Category=Star Description=[O stars]				
	#	Name	Criteria	Description	
	(3)	NIRSPec-TA-Generic	S/N>20 TA Star		

Proposal 12576 - Observation 1 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its ...

Observation	Proposal 12576, Observation 1: E1 Mon Jun 01 18:00:14 GMT 2026									
	Diagnostic Status: Error Observing Template: NIRCam Imaging									
Diagnostics	(E1 (Obs 1)) Error (Form): Minimum Response time is a required field.									
	(E1 (Obs 1)) Error (Form): Programs requesting Future Cycles may not contain a Disruptive Target of Opportunity.									
	(E1 (Obs 1)) Warning (Form): Response time < 14.0 Days is disruptive to the scheduling process. Only a limited number of disruptive ToOs will be accepted.									
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.									
	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
Template	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d)							
			Dec: -21 55 22.41 (-21.92289d)							
			Equinox: J2000							
			Comments: Category=Star Description=[Type Ia supernovae] Extended=NO							
Dithers	Module		Subarray							
	B		FULL							
Spectral Elements	#	Primary Dither Type	Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE			SMALL-GRID-DITHER				4	
Special Requirements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F115W	F444W	BRIGHT1	8	2	8	4	1331.359	
	2	F150W	F356W	BRIGHT1	8	2	8	4	1331.359	
	3	F200W	F277W	BRIGHT2	6	2	8	4	1073.677	
Special Requirements	Offset 55.0 arcsec, 45.0 arcsec On Hold On hold for SN Requiem detection Target Of Opportunity Response Time <None specified> to 7 Days 3 After 1 by 9 Days to 21 Days Group Observations 1, 2, Non-interruptible									

Proposal 12576 - Observation 3 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its ...

Observation	Proposal 12576, Observation 3: E2										Mon Jun 01 18:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 3:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d) Dec: -21 55 22.41 (-21.92289d) Equinox: J2000								
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				SMALL-GRID-DITHER				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT1	8	2	8	4	1331.359		
	2	F150W	F356W	BRIGHT1	8	2	8	4	1331.359		
	3	F200W	F277W	BRIGHT2	6	2	8	4	1073.677		
Special Requirements	Offset 55.0 arcsec, 45.0 arcsec On Hold On hold for SN Requiem detection										
	3 After 1 by 9 Days to 21 Days 4 After 3 by 9 Days to 21 Days										

Proposal 12576 - Observation 4 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its ...

Observation	Proposal 12576, Observation 4: E3										Mon Jun 01 18:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 4:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d)								
			Dec: -21 55 22.41 (-21.92289d)								
			Equinox: J2000								
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				SMALL-GRID-DITHER				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT1	8	2	8	4	1331.359		
	2	F150W	F356W	BRIGHT1	8	2	8	4	1331.359		
	3	F200W	F277W	BRIGHT2	6	2	8	4	1073.677		
Special Requirements	Offset 55.0 arcsec, 45.0 arcsec On Hold On hold for SN Requiem detection										
	4 After 3 by 9 Days to 21 Days 5 After 4 by 9 Days to 21 Days										

Proposal 12576 - Observation 5 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its ...

Observation	Proposal 12576, Observation 5: E4										Mon Jun 01 18:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 5:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d)								
			Dec: -21 55 22.41 (-21.92289d)								
			Equinox: J2000								
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				SMALL-GRID-DITHER				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT1	8	2	8	4	1331.359		
	2	F150W	F356W	BRIGHT1	8	2	8	4	1331.359		
	3	F200W	F277W	BRIGHT2	6	2	8	4	1073.677		
Special Requirements	Offset 55.0 arcsec, 45.0 arcsec On Hold On hold for SN Requiem detection										
	5 After 4 by 9 Days to 21 Days 6 After 5 by 9 Days to 90 Days										

Proposal 12576 - Observation 6 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its ...

Observation	Proposal 12576, Observation 6: E5										Mon Jun 01 18:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d) Dec: -21 55 22.41 (-21.92289d) Equinox: J2000								
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				SMALL-GRID-DITHER				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT1	8	2	8	4	1331.359		
	2	F150W	F356W	BRIGHT1	8	2	8	4	1331.359		
	3	F200W	F277W	BRIGHT2	6	2	8	4	1073.677		
Special Requirements	Offset 55.0 arcsec, 45.0 arcsec On Hold On hold for SN Requiem detection 6 After 5 by 9 Days to 90 Days 7 After 6 by 9 Days to 21 Days										

Proposal 12576 - Observation 7 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its ...

Observation	Proposal 12576, Observation 7: E6										Mon Jun 01 18:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d) Dec: -21 55 22.41 (-21.92289d) Equinox: J2000								
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				SMALL-GRID-DITHER				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT1	8	2	8	4	1331.359		
	2	F150W	F356W	BRIGHT1	8	2	8	4	1331.359		
	3	F200W	F277W	BRIGHT2	6	2	8	4	1073.677		
Special Requirements	Offset 55.0 arcsec, 45.0 arcsec On Hold On hold for SN Requiem detection 7 After 6 by 9 Days to 21 Days 8 After 7 by 9 Days to 21 Days										

Proposal 12576 - Observation 8 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its ...

Observation	Proposal 12576, Observation 8: E7										Mon Jun 01 18:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d)								
			Dec: -21 55 22.41 (-21.92289d)								
			Equinox: J2000								
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				SMALL-GRID-DITHER				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT1	8	2	8	4	1331.359		
	2	F150W	F356W	BRIGHT1	8	2	8	4	1331.359		
	3	F200W	F277W	BRIGHT2	6	2	8	4	1073.677		
Special Requirements	Offset 55.0 arcsec, 45.0 arcsec										
	On Hold On hold for SN Requiem detection										
	8 After 7 by 9 Days to 21 Days										
	9 After 8 by 9 Days to 21 Days										

Proposal 12576 - Observation 9 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its ...

Observation	Proposal 12576, Observation 9: E8										Mon Jun 01 18:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d)								
			Dec: -21 55 22.41 (-21.92289d)								
			Equinox: J2000								
Template	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO										
	Module						Subarray				
	B						FULL				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				SMALL-GRID-DITHER				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT1	8	2	8	4	1331.359		
	2	F150W	F356W	BRIGHT1	8	2	8	4	1331.359		
	3	F200W	F277W	BRIGHT2	6	2	8	4	1073.677		
Special Requirements	Offset 55.0 arcsec, 45.0 arcsec On Hold On hold for SN Requiem detection 9 After 8 by 9 Days to 21 Days										

Proposal 12576 - Observation 2 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its ...

Observation	Proposal 12576, Observation 2: E1-SPEC										Mon Jun 01 18:00:14 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(E1-SPEC (Obs 2)) Warning (Form): Observers are responsible for checking that target acquisition is feasible.											
	(E1-SPEC (Obs 2)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 43.139 Arcsec (larger than the recommended limit of 38.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d) Dec: -21 55 22.41 (-21.92289d) Equinox: J2000									
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	2 NIRSPec-TA	WATA	FULL	F110W	NRSRAPIDD6	3	1	1	171.788		
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	S200A1	NRSIRS2	11	4	1	NONE	5	20	16339.557	

Special Requirements	Group Observations 1, 2, Non-interruptible
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Proposal 12576 - Observation 10 - Requiem's Return: Precision cosmology from a decade-delayed, strongly-lensed supernova and its...

Observation	Proposal 12576, Observation 10										Mon Jun 01 18:00:14 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SN-R4	RA: 01 38 2.1602 (24.5090008d)								
			Dec: -21 55 22.41 (-21.92289d)								
			Equinox: J2000								
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				SMALL-GRID-DITHER				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F277W	SHALLOW4	7	1	4	4	1460.201		
	2	F150W	F356W	SHALLOW4	7	1	4	4	1460.201		
Special Requirements	Between Dates 14-DEC-2026:00:00:00 and 31-JAN-2027:00:00:00 Offset 55.0 arcsec, 45.0 arcsec										